



← NEWS AND ANNOUNCEMENTS

The Rockefeller Foundation Invests in Satellite Data & AI to Accelerate Economic Development and Climate Resilience in Africa

PRESS RELEASES

PUBLISHED DATE

August 29, 2022

New \$5.5 million collaboration with e-GUIDE and Atlas AI launches in Kenya, Nigeria, Rwanda, and Uganda to catalyze investment in agriculture, energy, and transportation infrastructure



Image: 2021 early release from collaboration examining AI analysis of human settlement activity in Nairobi.

NEW YORK / NAIROBI | August 29, 2022 — The Rockefeller Foundation announced a new \$5.5 million collaboration with e-GUIDE and Atlas AI to accelerate economic development and promote climate resilient infrastructure investment across sub-Saharan Africa. Leveraging satellite data and machine learning technologies, this three-year effort will produce unprecedented insight into the well-being of communities through a groundbreaking digital platform, which builds upon new research and publicly available data sets covering the nexus of agriculture, energy, and transportation sector development conditions. Initially covering Kenya, Nigeria, Rwanda, and Uganda, the platform will provide policymakers with extensive cross-sectoral insight into where new infrastructure development can help mitigate community vulnerabilities and promote economic opportunities, ultimately assisting efforts to prioritize and sequence investments more effectively in these key sectors.

“While data science has been used to improve individual development projects, we haven’t yet unlocked its potential to improve development at a systems level — which is critical, because efforts to drive change in energy, agriculture, and transportation must be integrated in order to make opportunity universal and sustainable,” said **Zia Khan, Senior Vice President of Innovation at The Rockefeller Foundation**. “We are excited about the potential of this collaboration to give policymakers, investors, and operators more dynamic situational awareness of local conditions and help them improve those conditions for the people they serve.”

Launched in 2018 by The Rockefeller Foundation alongside University of Massachusetts (UMass) Amherst, Columbia University, Carnegie Mellon University, **Rochester Institute of Technology**, and the Colorado School of Mines, e-GUIDE (Electricity Growth and Use In Developing Economies Initiative) is an effort to apply data science to electricity demand prediction in energy-poor emerging economies.

“We want to develop tools to measure how infrastructural developments such as roads, electricity systems, and agriculture lead to economic development,” said **Jay Taneja, Assistant Professor of Electrical and Computer Engineering at UMass Amherst, and**

principal investigator on the project. “We want to understand which combinations result in the most extensive and fastest economic development, primarily in countries in sub-Saharan Africa.”

e-GUIDE, which has successfully harnessed AI to predict electricity consumption in Africa and measure productive use of energy in Africa’s agriculture sector, is joined in the effort by Atlas AI, a public benefit technology startup based in Silicon Valley. Established in 2018 by The Rockefeller Foundation and a team of Stanford University professors, Atlas AI uses data from a range of planetary sensors along with deep learning AI technologies to monitor changes to global economic and societal wellbeing. Atlas AI has extensive experience building hyperlocal socioeconomic datasets, predictive analytics models, and software platforms to guide complex policy and investment decisions.

“For more than a decade, the scientific community has been demonstrating the use of data from new sensors like satellites along with AI technologies to measure specific development indicators such as the performance of staple crops, the wealth accumulation of households, and the pace of electrification,” said **Abe Tarapani, CEO of Atlas AI**. “Real-world investment decisions don’t fit as neatly into one of those buckets, and we are thrilled to be part of this pioneering consortium to unlock a more dynamic and practical understanding of how constrained resources can best promote livelihood improvements in local communities.”

e-GUIDE and Atlas AI have also partnered with the Kigali Collaborative Research Centre (KCRC) in Rwanda to leverage KCRC’s research and innovation leadership in energy systems, data science, artificial intelligence, transportation, and climate change across Africa.

Urgent Action Needed:

- Even though the share of Africa’s carbon emissions is less than 3% of global emissions, it is expected to bear the brunt of the economic impact of climate change according to the [African Development Bank](#).
- According to the [World Bank](#), current efforts to build resilience to climate change into Africa’s food systems are effective, but fall short when it comes to tackling the problem at scale.

- Hundreds of millions of people lack access to safe, reliable electricity in sub-Saharan Africa, hindering many governments' development and climate resilience goals.
- There is a lack of widespread data about Africa's transportation infrastructure constrains investment in resilient solutions in a sector that is responsible for around a quarter of all global emissions.

In response, the platform developed by e-GUIDE, Atlas AI, and KCRC is the result of thousands of hours of computation across petabytes of data covering electrification, agriculture, transportation, and human settlement activity, breaking down the silo-focused approach prevalent in many traditional sector-specific development data platforms.

"We're starting in Nigeria, Rwanda, Uganda, and Kenya," continued **Professor Taneja**, "because these four countries represent places where we already have strong relationships and are already collaborating on projects. We look forward to expanding the initiative's coverage to more countries in the coming years."

In addition, an advisory board is being created that will initially be composed of four independent experts with deep expertise in one or more sectors of interest.

About Atlas AI

Atlas AI is a predictive analytics platform that enables organizations to optimize performance and expand reach by investing in underserved communities around the world. Founded by a team of Stanford University scientists in partnership with The Rockefeller Foundation and with support from Airbus Ventures and Micron Technologies Inc., we harness a range of planetary scale datasets and the latest advances in artificial intelligence to measure local socioeconomic change such as population demographics, infrastructure access, household spending power and crop yields. We use this proprietary data to analyze, monitor and forecast regions of growth and untapped potential to offer insight into where organizations can grow most successfully, and where investment can best drive economic progress. Visit <http://atlasai.co>.

About e-GUIDE

The Electricity Growth and Use in Developing Economies (e-GUIDE) Initiative seeks to transform the approaches used for planning and operations of electricity infrastructure in developing regions. We are constructing measurement and data analytics techniques that are scalable, transnational, and verified using real data on electricity consumption and infrastructure. We partner with electricity service companies to develop our techniques, deploy them at scale, and build capacity for data and analytics in the electricity sector. The Initiative is funded through \$5.8 million in grants from The Rockefeller Foundation. Find out more: <https://eguide.io/>

About The Rockefeller Foundation

The Rockefeller Foundation is a pioneering philanthropy built on collaborative partnerships at the frontiers of science, technology, and innovation to enable individuals, families, and communities to flourish. We work to promote the well-being of humanity and make opportunity universal. Our focus is on scaling renewable energy for all, stimulating economic mobility, and ensuring equitable access to healthy and nutritious food. For more information, sign up for our newsletter at rockefellerfoundation.org and follow us on Twitter [@RockefellerFdn](https://twitter.com/RockefellerFdn).

Media Contacts

Ashley Chang

Head of Media Relations

+1 212 852 0000

achang@rockfound.org

Umuliza M. Njiru

Lead, Communications

Policy & Advocacy for Africa

+254 20 4987517

unjiru@rockfound.org

Stay Updated

Get quick updates in our e-newsletter.

SUBSCRIBE NOW →



Related Updates





BLOG POST

Another Energy Crisis Is Here. This Time, the Way Out Is Different.



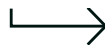
PRESS RELEASES

Mission 300 Launches Private Sector Council to Boost Electricity Access and Job Creation in Africa



BLOG POST

How Food is Medicine Can Strengthen Farms, Communities, and Health

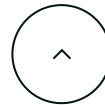


SHOW MORE UPDATES

Tags :

ARTIFICIAL INTELLIGENCE

INNOVATION



Back to Top



Resources

[Our Grants](#)

[Financial Documents](#)

[Privacy Policy](#)

Locations

[All Offices](#)

[New York City, United States](#)

[Washington, D.C., United States](#)

- Terms of Use
- Accessibility Commitment Statement
- Transparency in Coverage Rule

- Bellagio Center, Italy
- Nairobi, Kenya
- Bangkok, Thailand
- Bogotá, Colombia

Get In Touch

- Contact Us
- Careers

The Rockefeller Foundation's mission is to promote the well-being of humanity and make opportunity universal and sustainable.

Subscribe

Sign up for our newsletter to stay updated on all the latest news and events.

SIGN UP →

f

✕

in

@

▶

Copyright 2026 The Rockefeller Foundation.
All Rights Reserved

